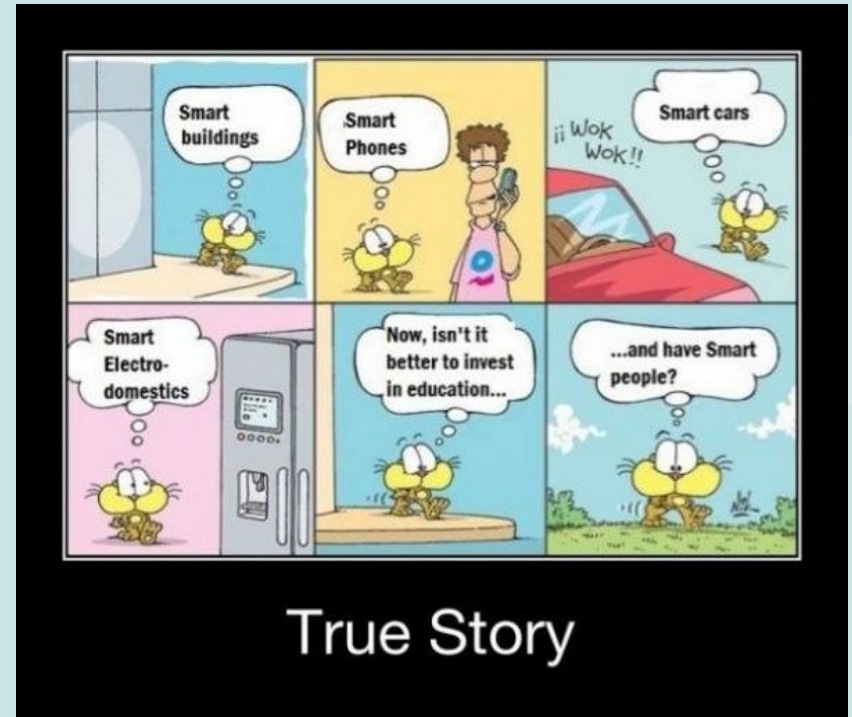


QA and TEST

Business facing test



Agenda

1. Status på softwarekvalitet-faget.
2. Moq undersøgelse
3. Business facing test intro (Q2)
4. ReqnRoll
5. Hjemmeopgave (I projektet)



Hvad har vi lært?



Hvad har vi lært?

Vi har bla. lært følgende.

1. Solid principperne (Q1)
 - a. Struktur af kode der skal testes
2. Unit testing (Q1)
 - a. Struktur
 - b. Testmiljø
 - c. Fluent assertion
3. Moq (Q1)
 - a. Hurtigere test (formål og udførelse)

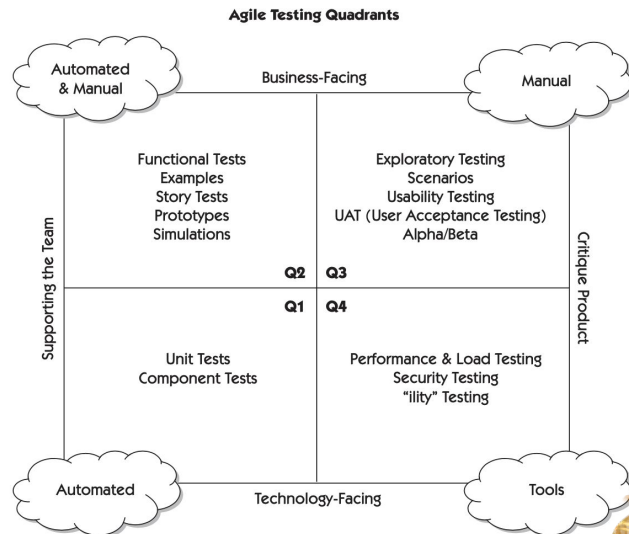


Figure 6-1 Agile Testing Quadrants





Mocking



Business facing tests

and Reqnroll

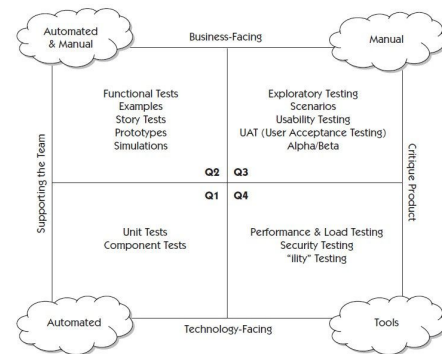
Business facing tests

Q2 repræsentere funktionstest og story tests m.m.

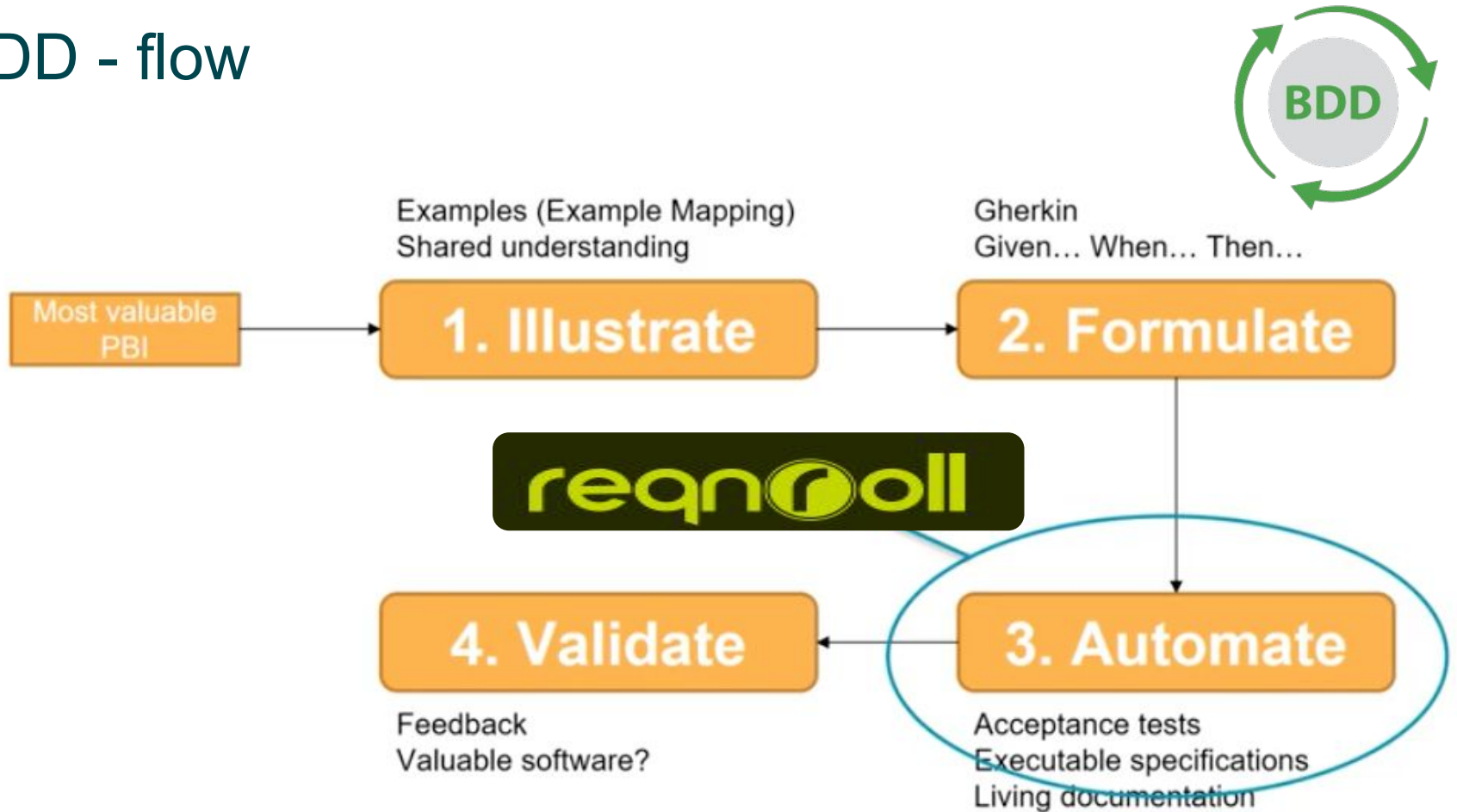
I Q1 lavede vi unit tests for at teste om koden der er lavet gør det som udvikleren selv forventer. Når vi rykker til Q2 er spørgsmålet nu om udvikleren har forstået opgaven korrekt.

Kan man på et mindre komplekst niveau teste forståelsen af opgaven og bruge automatisering af lavt niveau tests?

ReqNRoll giver os en mulighed for dette.



BDD - flow



BBD explained



ReqnRoll (fra bogen men opdateret)



ReqnRoll is the #1 .NET open source framework for BBD, Acceptance Test (not user AT) Driven Development and Specification by Example. With over 10m downloads on NuGet, ReqnRoll is trusted by teams around the world

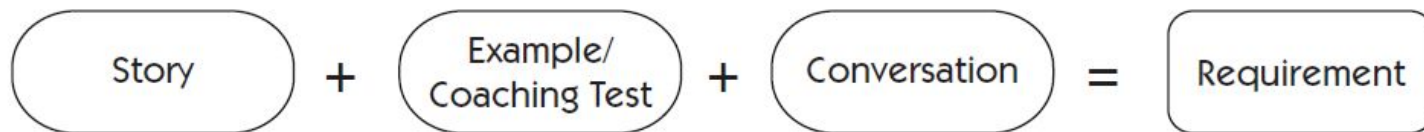


Figure 8-3 The makeup of a requirement

Types of general questions to ask in BDD

- Is this story solving a problem?
- If so, what's the problem we're trying to solve?
- Could we implement a solution that doesn't solve the problem?
- How will the story bring value to the business?
- Who are the end users of the feature?
- What value will they get out of it?
- What will users do right before and right after they use that feature?
- How do we know we're done with this story?

What's the worst thing that could happen?

Worst-case scenarios tend to generate ideas.



Stories that look small but that impact unexpected areas of the system can come back to bite you

Flowdiagram

Flowdiagram er et fornuftigt værktøj til at beskrive funktionalitet, men kan Kunden/PO forstå det?

LoFi wireframe.

LoFi functional wireframe.

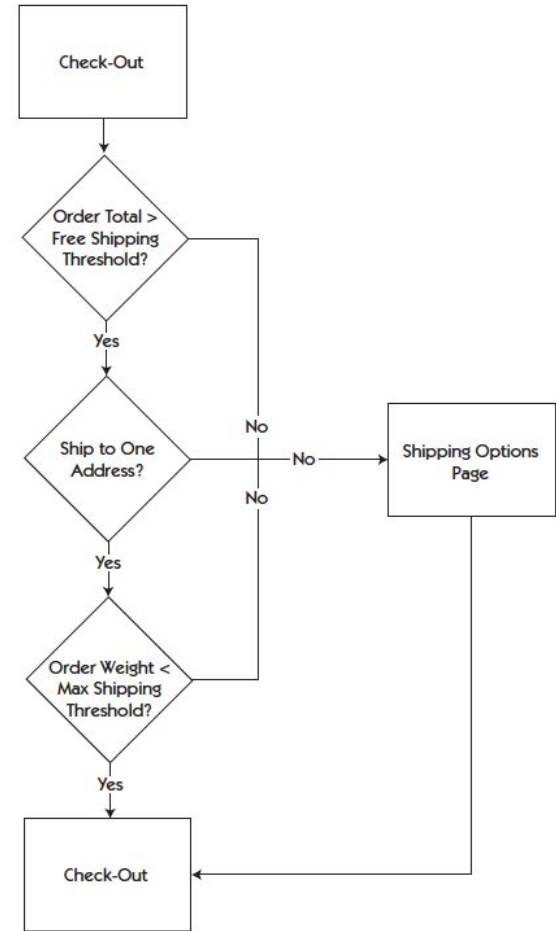
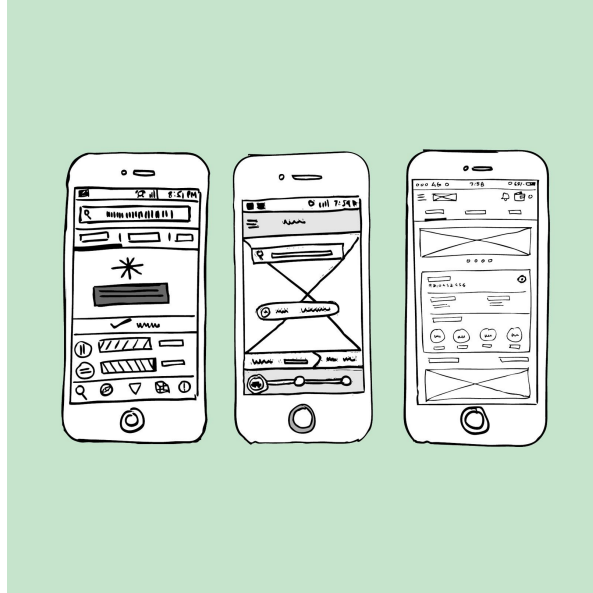


Figure 9-7 Flow chart for qualifying for free shipping option

OPG 1: Opsætning af ReqnRoll

Sæt ReqnRoll op. (for dem der ikke kommer forberedt)

Der er hjælp at hente på ReqnRoll website.

<https://reqnroll.net/>

Gennemfør quickstart guide:

<https://docs.reqnroll.net/latest/quickstart/index.html>

Hvis du er færdig før andre i teamet, så hjælp de resterende.

Story

Stakeholder: For the next release of our online store, our Goldlevel customers should receive a discount when they make a purchase.

Developer: What kind of discount—what criteria do they have to meet in order to receive it?

Stakeholder: When they have at least \$50 dollars in their shopping cart.

Developer: Does the discount increase based upon the amount, or is it fixed regardless of the value of the shopping cart?

Stakeholder: Good question—the discount is fixed at 15% regardless of price. So, given a Gold-level customer, when the shopping cart totals \$50 or more, it should receive a 15% discount off the total price.

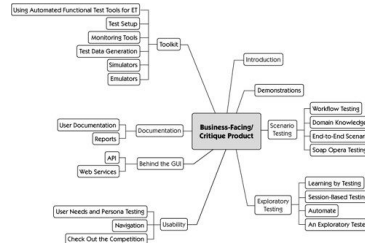
Bog: AT s. 167

Business facing test eksempel



scenario “Gold-level customer with \$50 in shopping cart”, {
given “ a Gold-level customer”
when “their shopping cart totals \$50 or more”
then “ they should receive a 15% discount off the total price”
}

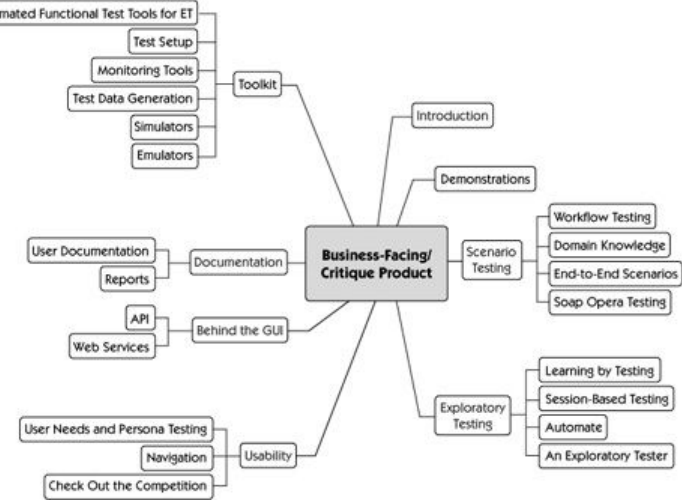
Bog: AT s. 167



Business facing test eksempel v2

```
scenario "Gold-level customer with $50 in shopping cart", {  
  given "a Gold-level customer",  
  {customer = new GoldCustomer()}  
  when "their shopping cart totals $50 or more",  
  {customer.shoppingCart << new Item("widget", 50.00)}  
  then "they should receive a 15% discount off the total price" ,  
  {customer.orderPrice.shouldBe 42.50}  
}
```

Bog: AT s. 167



Example

Feature: Basket Functionality

As a user

I want to be able to add items to my basket

So that I can keep track of the total price of my basket contents

Scenario: Add Two Items to Basket and Check Total Price

Given I have an empty basket

And the price of a "Book" is \$10.00

And the price of a "DVD" is \$5.00

When I add a "Book" to the basket

And I add a "DVD" to the basket

And I add a "DVD" to the basket

Then the total price of the basket contents should be \$20.00

And the total price of the "Book" in the basket should be \$10.00

And the total price of the "DVD" in the basket should be \$10.00

Gherkin - Background

Feature: Multiple site support

Only blog owners can post to a blog, except administrators, who can post to all blogs.

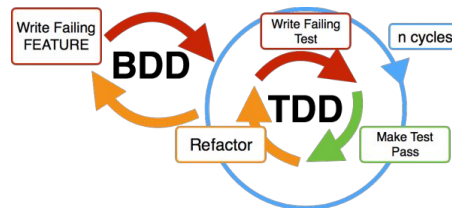
Background:

Given a global administrator named "Greg"

And a blog named "Greg's anti-tax rants"

And a customer named "Dr. Bill"

And a blog named "Expensive Therapy" owned by "Dr. Bill"



Scenario: Dr. Bill posts to his own blog

Given I am logged in as Dr. Bill

When I try to post to "Expensive Therapy"

Then I should see "Your article was published."

Scenario: Dr. Bill tries to post to somebody else's blog, and fails

Given I am logged in as Dr. Bill

When I try to post to "Greg's anti-tax rants"

Then I should see "Hey! That's not your blog!"

Scenario: Greg posts to a client's blog

Given I am logged in as Greg

When I try to post to "Expensive Therapy"

Then I should see "Your article was published."

<https://cucumber.io/docs/gherkin/reference/#background>

Gherkin - Rule

Feature: Overdue tasks

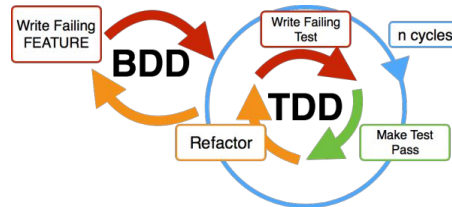
Let users know when tasks are overdue, even when using other features of the app

Rule: Users are notified about overdue tasks on first use of the day

- **Background:**

Given I have overdue tasks

Whilst usage of Background within Rule is currently supported, it's not recommended, and may be removed in future versions of Gherkin



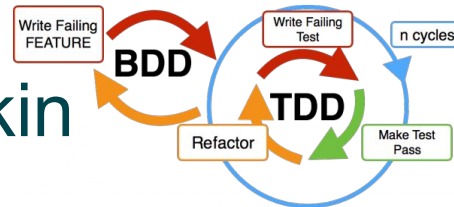
Example: First use of the day

Given I last used the app yesterday
When I use the app
Then I am notified about overdue tasks

Example: Already used today

Given I last used the app earlier today
When I use the app
Then I am not notified about overdue tasks

Tips for using background in Gherkin



- Don't use Background to set up complicated states, unless that state is actually something the client needs to know.
- For example, if the user and site names don't matter to the client, use a higher-level step such as Given I am logged in as a site owner.
- Keep your Background section short.
- The client needs to actually remember this stuff when reading the scenarios. If the Background is more than 4 lines long, consider moving some of the irrelevant details into higher-level steps.
- Make your Background section vivid.
- Use colourful names, and try to tell a story. The human brain keeps track of stories much better than it keeps track of names like "User A", "User B", "Site 1", and so on.
- Keep your scenarios short, and don't have too many.
- If the Background section has scrolled off the screen, the reader no longer has a full overview of what's happening. Think about using higher-level steps, or splitting the *.feature file.

Open Source BDD TOOLS



cucumber



Behavior Driven Development & Automation Testing



Write failing Test



Implement features



Ensure that tests pass



Automated Testing



Using ReqNRoll testing API

Exercise semesterprojekt

1. Prøv at lave en/nogle test til semesterprojektet i Gherkin
2. Hvis din kode er “clean” vil Implementering til de test der er lavet i punkt 1 kunne laves direkte på koden.





API testing

With reqNroll - vi laver et fælles miniprojekt - i undervisningen

Setup github API access



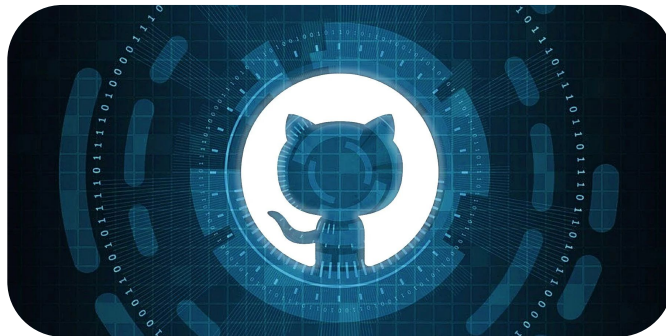
1. Setup your github so you can access it by following this guide:
<https://docs.github.com/en/rest/using-the-rest-api/getting-started-with-the-rest-api?apiVersion=2026-03-10>
2. Tjek om det latest version.



Kode hjælp til github kald i C# uden authorization

(2024)

```
client.DefaultRequestHeaders.Accept.Clear();  
client.DefaultRequestHeaders.Accept.Add(  
    new MediaTypeWithQualityHeaderValue("application/vnd.github.v3+json"));  
client.DefaultRequestHeaders.Add("User-Agent", ".NET Foundation Repository Reporter");  
  
try  
{  
    string response = await client.GetStringAsync(url);
```



Feature: Get a list of issues - 1st try



Scenario: I want to get a list of the issues for the Symfony repository

Given I am an anonymous user

When I request a list of issues for the Symfony repository from user Symfony

Then I should get at least 1 result

<https://github.com/symfony/symfony>

Feature: Get a list of issues - using variables

Scenario: I want to get a list of the issues for the Symfony repository

Given I am an anonymous user

When I request a list of issues for the “Symfony” repository from user “Symfony”

Then I should get at least 1 result



Postman:

Prøv følgende:

<https://api.github.com/repos/Symfony/Symfony/issues>

Dette skulle gerne give en status 200 tilbage, hvilket betyder at “WE ROCK”



Præsentation af 1. del af opgaven

hvem er klar?

Brugeradgang



1. setup access token

<https://www.softwaretestinghelp.com/github-rest-api-tutorial/>

video:

<https://www.linkedin.com/learning/api-testing-and-validation-22610641/expand-your-testing-environment?resume=false&u=57075649>

2. Test det af med postman
3. Goto github and create a repository called "is-your-api-misbehaving"

Get my list of repositories

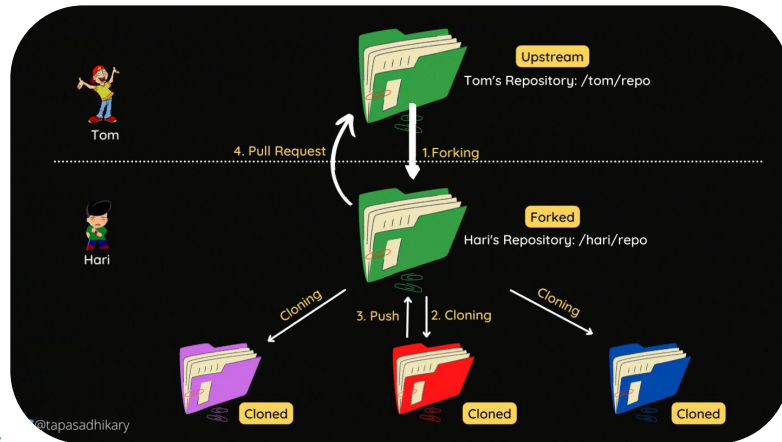
Feature: Get my list of repositories

Scenario: I want to find my private repository

Given I am an authenticated user

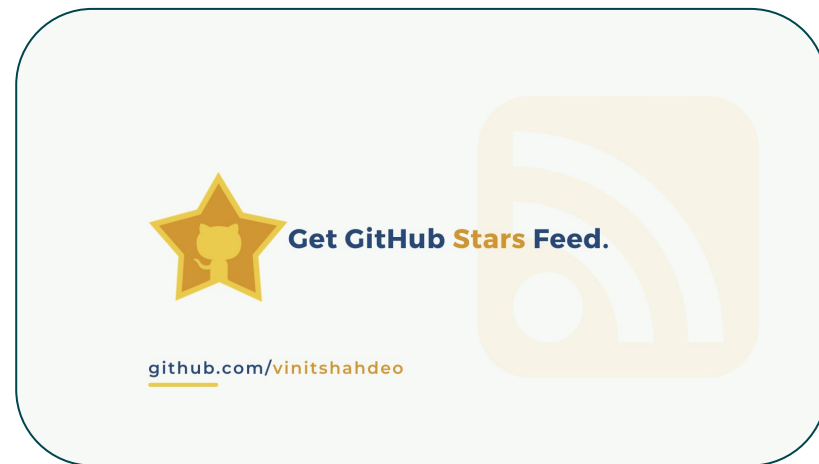
When I request a list of my repositories

Then the results should include a repository named "is-your-api-misbehaving"



My first write to github API

Feature: Star a repository



Scenario: I want to star an important repository

Given I am an authenticated user

When I star my "is-your-api-misbehaving" repository

Then my "is-your-api-misbehaving" repository will list me as a stargazer

Cleaning up after test while testing opposite behavior

Feature: Unstar a repository

Scenario: I want to unstar an important repository

Given I am an authenticated user

When I unstar my "is-your-api-misbehaving" repository

Then my "is-your-api-misbehaving" repository will not list me as a stargazer



Create a new repository

Feature: Create a repository

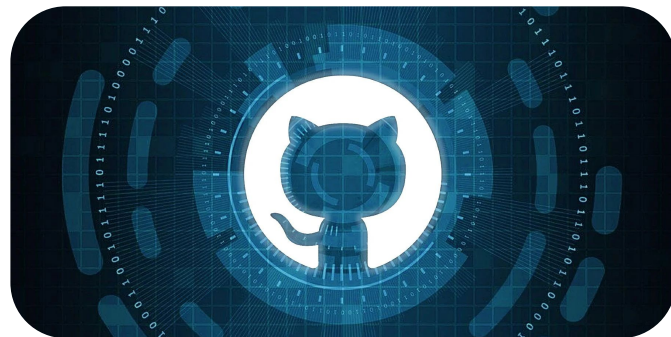
Scenario: I want to create a repository

Given I am an authenticated user

When I create a repository called "something-cool"

And I request a list of my repositories

Then the results should include a repository named "something-cool"



Clean and test delete functionality

Feature: Delete a repository

Scenario: I want to delete a repository



Given I am an authenticated user

And I request a list of my repositories

And the results should include a repository named "something-cool"

When I delete a repository called "something-cool"

And I request a list of my repositories

Then the results should not include a repository named "something-cool"

Exercise semesterprojekt API test

1. Lav test til API i ReqNRoll (semesterprojekt api er foretrukket)



Hjemmeopgave



opg 1. Opsætning af ReqnRoll i projektet, test af API i projektet via reqnroll. (ikke nødvendigvis alle funktionaliteter)

Opg 2 Gør projektet klar til midtvejsevaluering dvs.at det kan vises til mig næste gang selvom det ikke er færdigt.

